

Fine-Tuning a VLM for Report Generation and VQA

A simple, practical plan for fine-tuning one vision-language model (VLM) to both write chest X-ray reports and answer visual questions.

1. Goal

Train a single VLM that can do two tasks from a chest X-ray image:

- Report generation: write a radiology-style report (Findings + Impression).
- Visual question answering (VQA): answer natural-language questions about the image.

Both tasks share the same pattern: image + instruction produces text. Only the prompt and the target text change per task.

2. Task objectives

Report generation

- Input: chest X-ray image (optionally view and indication metadata).
- Output: radiology-style text with Findings and Impression sections.
- Success: correct pathology mentions, no hallucinated findings, readable structure.

Visual question answering

- Input: image plus a natural-language question.
- Output: short answer (yes or no, a label, or one to two sentences).
- Success: high accuracy on closed questions, sensible answers on open questions.

3. Data required

Minimum fields per sample

Field	Report generation	VQA
image_path	required	required
target_text	full or sectioned report	ground-truth answer
question	not used	required
split	train / val / test	train / val / test
patient_id	recommended	recommended

Report generation datasets

Dataset	Contents	Notes
MIMIC-CXR	~377k images, ~227k reports	Best for report gen; needs PhysioNet access
IU X-Ray	~7k images and reports	Small, good for prototyping
PadChest	Reports plus labels	Scale and diversity (Spanish)

CheXpert alone is not enough for report generation: it has labels, not narrative reports.

VQA datasets

Dataset	Type	Notes
VQA-RAD	Radiology VQA	Small (~3k questions), good starter
SLAKE	Medical VQA incl. CXR	Bilingual
RadVQA	Large radiology VQA	Built from reports
MIMIC-CXR-VQA	Synthetic Q&A from MIMIC	Pairs well if using MIMIC

Optional data

- Findings-only labels (CheXpert, NIH) for weak or auxiliary supervision.
- Bounding boxes or segmentation for location questions.
- Synthetic Q&A generated from existing reports with an LLM to scale VQA cheaply.

4. Data format (shared SFT schema)

Every sample becomes a chat turn: user provides image plus instruction, assistant provides the target text.

Report example row:

```
{
  "task": "report",
  "path": "images/study_001.png",
  "prompt": "Write a chest X-ray report with FINDINGS and IMPRESSION.",
  "target": "FINDINGS: ... IMPRESSION: ..."
}
```

VQA example row:

```
{
  "task": "vqa",
  "path": "images/study_001.png",
  "prompt": "Question: Is there pleural effusion?",
  "target": "Yes"
}
```

Use a task prefix in the prompt so the model knows the expected output format:

```
[TASK: REPORT] Write findings and impression for this chest X-ray.
[TASK: VQA] Answer the following question about this chest X-ray.
```

5. Training both tasks in one model

Recommended: mixed multi-task SFT with a single shared LoRA adapter.

- Put report and VQA samples in one training file and shuffle each epoch.
- Starting mix: 60-70% report generation, 30-40% VQA.
- If VQA lags, oversample VQA to ~50% or run a short VQA-only stage afterward.

Alternative: staged training.

1. Fine-tune on reports first (learn radiology language).
2. Continue LoRA on VQA with a smaller learning rate (~5e-6).
3. Optional short mixed fine-tune at the end.

6. Training recipe

- Base model: Qwen3.5-4B or Qwen2-VL-2B-Instruct (instruct/chat variant required).
- Method: LoRA (rank 16-32) on attention and MLP projection layers.
- Learning rate: $\sim 1e-5$ to $2e-5$ (use $\sim 5e-6$ for a second staged step).
- Batch size 1-4 with gradient accumulation 8-16.
- Epochs: 1-3, watching validation loss (radiology data overfits fast).
- Enable gradient checkpointing if GPU memory is tight.
- Split by patient or study, never by image, to avoid leakage.

7. Evaluation (keep tasks separate)

Task	Metrics
Report generation	BLEU, ROUGE-L, METEOR; RadGraph F1 or CheXbert label F1; manual read of 50-100 samples
VQA	Accuracy (closed); exact match or token F1 (open); per-question-type breakdown

Always evaluate each task independently. Run the same prompts on the base model before LoRA to establish the improvement baseline.

8. Fit with the existing repo

The current stack already fine-tunes Qwen with LoRA SFT for JSON label prediction

```
(`scripts/train_qwen35_lora_sft.py`, `build_sft_batch` in
`scripts/qwen2vl_lora_common.py`).
```

To extend it for report and VQA:

- Generalize ``target_json_from_row`` to ``target_text_from_row(row)``, where ``row["task"]`` selects the report or VQA prompt and target.
- Point training at the new mixed report+VQA rows file.
- Add report and VQA evaluation metrics alongside the existing macro-F1.

One LoRA, one training run, mixed batches.

9. Suggested roadmap

1. Week 1: VQA prototype on VQA-RAD or SLAKE; get accuracy on a test set.
2. Week 2: Report generation on IU X-Ray or a MIMIC subset; RadGraph/CheXbert eval.
3. Week 3: Combine into mixed SFT ($\sim 70\%$ report / 30% VQA) with task prefixes.
4. Week 4: Clinical error analysis (hallucinations, missed findings, view confusion).

10. Summary

- Objective: one VLM that describes the image (report) and answers questions (VQA).
- Data: image plus report text, and image plus question plus answer, with patient-level splits.
- Method: LoRA SFT in chat format, mixed multi-task data with clear per-task prompts.
- Evaluation: report metrics and VQA metrics measured separately.